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List of Exhibits

The eight data sets analysed in Chapter 4 are supported by the following tables and figures. Each data set follows the same four-part structure: the table or graph, an explanation of the data, an analysis, and an interpretation.

Exhibit	Title	Data set
Table 1	Growth of UPI transactions, FY 2016-17 to FY 2025-26	Data Set 1
Figure 1	UPI transaction volume, FY 2016-17 to FY 2025-26	Data Set 1
Figure 2	UPI transaction value, FY 2016-17 to FY 2025-26	Data Set 1
Table 2	Average value of a single UPI transaction	Data Set 2
Figure 3	Average value of a single UPI transaction	Data Set 2
Table 3	UPI and total digital payments compared	Data Set 3
Figure 5	UPI's share of retail digital payment volume	Data Set 3
Table 4	UPI turnover relative to nominal GDP	Data Set 4
Figure 4	UPI turnover relative to nominal GDP	Data Set 4
Table 5	Digital payments around demonetisation	Data Set 5
Figure 6	Digital payments around demonetisation	Data Set 5
Table 6	UPI growth before, during and after COVID-19	Data Set 6
Figure 7	UPI volume before, during and after COVID-19	Data Set 6
Table 7	Indicators of financial inclusion and digitalisation	Data Set 7
Figure 8	Financial inclusion and payment digitalisation indices	Data Set 7
Figure 10	India's share of global real-time payment volume	Data Set 7
Table 8	Internet and telecom penetration, rural and urban	Data Set 8
Figure 9	The rural-urban digital divide	Data Set 8

1. Introduction

1.1 The meaning of digital payments

A payment means handing over money to settle what you owe. For most of history this needed a physical object to change hands: a coin, a note or a cheque. A digital payment removes the object. The money moves as a message instead. That message passes between two bank accounts, the payer's balance falls, the payee's balance rises, and no cash is touched at any stage.

This difference matters more than it looks. Cash settles at once, but it moves slowly. It has to be withdrawn, carried, counted and deposited again. A digital instruction travels at the speed of a phone network and settles in seconds. So what changes is not just how easy it is to pay. What also changes is how many times the same money can be used in a year. That idea is the thread this project follows from start to finish.

1.2 The Unified Payments Interface

The Unified Payments Interface, known as UPI, was launched in India in 2016. It is not an app. It is a standard, which means a common set of rules. Those rules let any bank account be paid from any app, using a single address instead of an account number. A user of one app can pay a shopkeeper who banks elsewhere and uses a completely different app. The payment is instant, free for both sides, and works at any hour.

Three features explain why UPI worked where earlier systems did not. First, it is interoperable, which means the payer and the payee do not have to use the same provider. Second, it never shuts down, unlike the older systems it replaced. Third, it is free for ordinary payments. The Government has supported this by keeping the Merchant Discount Rate at zero for small-merchant UPI payments up to ₹2,000. In place of that fee it pays the banks an incentive of 0.15 per cent of the transaction value [14].

The International Monetary Fund studied this design in June 2025 and said it was the main reason UPI spread so fast. Interoperability gives users the freedom to choose their app. Many people join through a brand they already trust, then move to a better app later. Because users can leave at any time, every provider has to keep improving. The result was a market with more than 200 apps, rather than one company taking everything [18].

1.3 The growth of digital transactions in India

The scale of the change is hard to describe without figures. In FY 2016-17, its first full year, UPI handled about 2 crore transactions worth ₹0.07 lakh crore. In FY 2025-26 it handled over 24,162 crore transactions worth about ₹314 lakh crore. That is roughly a twelve-thousand-fold rise in the number of payments and a more than four-thousand-fold rise in their value, in ten years [2].

This is not a small or specialist service. UPI handled 81 per cent of all retail digital payments in India in FY 2024-25 [14]. The IMF calls it the largest retail fast payment system in the world by number of transactions. It carries close to 49 per cent of all real-time payments made anywhere on earth [3], [18]. By June 2026, 55.49 crore users had joined [16]. The number of banks on the platform rose from 44 to 703 by FY 2025-26 [2].

1.4 The role of the National Payments Corporation of India

None of this was left to the market to arrange. UPI was built and is run by the National Payments Corporation of India (NPCI). NPCI is the umbrella body for retail payments, set up under the guidance of the Reserve Bank of India and the Indian Banks' Association. It is registered as a not-for-profit company, and that matters for the economics. Because NPCI does not have to earn a profit on the system itself, payments can be carried at or near cost.

NPCI does five main jobs. It writes the technical rules. It runs the central switch that every payment passes through. It settles the amounts owed between member banks. It sets and revises transaction limits. And it runs the systems that watch for fraud as payments happen. Through its subsidiary NPCI International Payments Limited, it has also taken UPI abroad. UPI now works in eleven foreign countries for payments or money transfers, including the United Arab Emirates, Singapore, Bhutan, Nepal, Sri Lanka, France, Mauritius and Qatar [17].

The fact that this is public infrastructure has drawn attention abroad. The Bank for International Settlements calls systems like UPI digital public infrastructure. By this it means open systems that anyone can connect to and that support services used by the whole of society. It notes that Aadhaar and UPI together have made large gains in financial inclusion and in the efficiency of payments [20].

1.5 From a cash economy to a digital economy

India began this decade holding more cash, relative to the size of its economy, than most countries. The Reserve Bank has treated the reduction of that ratio as a goal of policy and named it in Payments Vision 2025 [26]. Cash is not free. It has to be printed, moved, guarded, sorted and destroyed. More importantly, every rupee sitting in a pocket or a cash box is a rupee doing no work.

The move away from cash has therefore been pushed on purpose, not left to convenience. A survey commissioned by the Department of Financial Services, released in February 2026, found that 90 per cent of users felt more confident about digital payments after using UPI and RuPay. The same survey recorded a clear fall in their use of cash and in ATM withdrawals [15]. This is what the national figures look like at the level of one household.

1.6 The meaning of the velocity of money

The velocity of money is the average number of times one rupee is used to buy goods and services in a given period. It is not a physical quality of money. It describes how quickly people spend the money they hold.

Irving Fisher set this out in his equation of exchange, usually written as:

$$M \times V = P \times T$$

Here M is the stock of money in circulation, V is the velocity of that money, P is the average price level and T is the number of transactions. The equation says something simple. If the stock of money stays the same and the number of transactions rises, then velocity must have risen. In its more common income form the equation becomes $V = (P \times Y) \div M$, where $P \times Y$ is national income at current prices.

It is important to say which velocity is meant, and this project is careful about that. Income velocity compares national output with the money stock. It moves slowly. Transactions velocity compares the value of all payments with the money stock, and it can move much faster. The difference is that transactions velocity counts every payment, including transfers that are not purchases of anything newly produced. Payment systems affect transactions velocity directly and income velocity only indirectly. Section 3.4 explains this at length, because mixing up the two is the most common mistake made on this subject.

1.7 Digital payments and economic activity

How could a payment system change real economic activity? Four routes are usually suggested, and each one appears again in the research reviewed in Chapter 3.

1. The cost of paying falls. When a payment costs nothing in fees and almost nothing in time, exchanges that were not worth the trouble before start to happen. This matters most for very small payments, which cash handled badly and cards never reached at all.
2. People hold less idle money. A household that can pay instantly from its account does not need to keep much cash in hand. Money held for spending

falls compared with the amount actually spent, and that is the same thing as a rise in velocity.

3. The informal economy becomes visible. A digital payment leaves a record. Records build up into a credit history, and a credit history can turn a trader with no bank loan into a borrower. The Reserve Bank has pursued this directly through the Unified Lending Interface [20].
4. Working capital is reused faster. A shopkeeper paid at once, rather than at the end of the day, can buy new stock sooner. The same rupee of working capital then pays for more sales in a year.

These are reasonable explanations, not measured amounts, and this project treats them that way throughout.

1.8 Financial inclusion and the formalisation of the economy

Financial inclusion means being able to use financial services that are useful and affordable. India built this in layers, and each layer needed the one before it. First came bank accounts for everyone under the Pradhan Mantri Jan Dhan Yojana. Then came a digital identity to prove who the account holder was. Last came a payment system those accounts could actually use. By 4 August 2025 the Jan Dhan scheme had reached over 55.98 crore people, and more than 55 per cent of those accounts were held by women [7].

The result can be measured. The Reserve Bank's Financial Inclusion Index rose from 64.2 in March 2024 to 67.0 in March 2025, and all three of its parts improved: access, usage and quality [7]. Formalisation comes from usage, not from access alone. An account that is only opened changes nothing. An account that is used creates the record of transactions on which credit and insurance can later be built.

1.9 Relevance in the post-demonetisation and post-pandemic economy

Two shocks mark UPI's first ten years. In November 2016, notes of ₹500 and ₹1,000 stopped being legal tender. They were worth ₹15.4 trillion, which was 86.9 per cent of the value of all notes then in circulation [23]. Monthly digital transactions rose 56 per cent between October 2016 and May 2017 [24]. UPI had launched only months earlier, so it was available at the exact moment the country was forced to look for something other than cash.

The second shock was COVID-19. It turned contactless payment into a health precaution rather than a convenience, and it pushed a whole generation of small shopkeepers into accepting a QR code. The two episodes worked differently, and

Data Sets 5 and 6 examine that difference. The first was a sharp force that people partly reversed afterwards. The second was a slower change that did not reverse.

1.10 Statement of the problem

Nobody disputes that UPI has grown. The figures are published every month and are not seriously questioned. The question this project asks is narrower and harder. Has that growth changed the speed at which money circulates in India? And can any effect on real economic activity be found in data that is publicly available?

The gap between those two questions is what this report is really about. A payment system can move a huge amount of money without changing national income at all. That happens if all it does is shift payments that would have been made in cash anyway. To show that something more than shifting has happened, different evidence is needed: evidence about the size of transactions, about who is paying, and about what shopkeepers report. That evidence is collected in Chapter 4.

2. Design of Study

2.1 Objectives of the study

This study has five objectives.

5. To understand what UPI is, how it is built, and how it grew in India between FY 2016-17 and FY 2025-26.
6. To examine the link between UPI payments and the velocity of money, keeping transactions velocity and income velocity clearly separate.
7. To study the effect of digital payments on economic activity, including spending patterns, the sales of small shopkeepers, and the recording of transactions.
8. To examine how far UPI has helped financial inclusion and brought informal economic activity into the recorded economy.
9. To study the problems and limits of digital payment systems in India, including cyber fraud, dependence on infrastructure, and uneven adoption across regions.

2.2 Research questions and hypotheses

These objectives are turned into three statements that can be tested. They are given here so that the reader can judge, at the end of Chapter 4, whether the evidence supports them.

	Hypothesis	Evidence used
H1	The value of payments settled for every rupee of national output has risen sharply since FY 2016-17. In other words, transactions velocity has increased.	Data Sets 1 and 4
H2	UPI's growth comes from genuinely new, small payments, and not only from large payments moving across from cash.	Data Sets 2 and 3
H3	The gains in inclusion and usage are not shared evenly between rural and urban India.	Data Sets 7 and 8

2.3 Data collection methodology

This study uses secondary data only. No survey was carried out, for a simple reason. The subject is national in scale, and it is already measured in full, every month, by the bodies that run and regulate it. A survey done by a school student could not improve on those measurements. Any sample small enough to be practical would be far too small to represent the country.

Sources were chosen in this order of preference:

- Operator and regulator data. Transaction statistics from the National Payments Corporation of India, and the Annual Report, Payment Systems Report, Digital Payments Index and Financial Inclusion Index of the Reserve Bank of India.
- Government releases. Press Information Bureau releases and factsheets, answers given in Parliament, and the Economic Survey.
- Official statistics. National income figures from the Ministry of Statistics and Programme Implementation, and telecom and internet data from the Telecom Regulatory Authority of India.
- International research. Publications of the International Monetary Fund, the Bank for International Settlements and the World Bank, used mainly for comparison and for guidance on method.
- Academic research. Peer-reviewed papers and pre-prints, used where they show things at the level of individual users that national figures cannot.

Commercial and news websites were avoided as sources of figures. Every number quoted here comes from the body that produced it. The number in square brackets points to the entry in the Bibliography where that release can be found.

Two measures in this report were worked out by me, rather than taken from a source. The first is the average value of a UPI transaction, found by dividing yearly value by yearly volume. The second is UPI turnover as a multiple of nominal GDP. Both methods are written out next to the relevant tables so that anyone can check them.

2.4 Scope of the study

This study looks at the effect of digital payments on the Indian economy as a whole. Within that, it focuses on UPI. Other instruments such as cards, wallets, IMPS, NEFT and RTGS appear only where a comparison is needed. The period runs from FY 2016-17, UPI's first full year, to FY 2025-26, the most recent year with complete data when this was written. The main ideas examined are the velocity of money, consumption, and the recording of financial activity.

Some related questions are left out on purpose. This study does not look at the profits of payment companies, at the design of a central bank digital currency, or

at the competition question of how few apps handle most UPI traffic. Each of those is a large subject in itself.

2.5 Limitations of the study

The limits below are stated openly. A study that hides them asks for more trust than its evidence deserves.

10. The study uses secondary data. Its conclusions can only be as reliable as the published figures they rest on.
11. Correlation is not causation. UPI grew during a decade in which smartphone ownership, internet access, bank accounts and incomes all rose together. This project can show that things moved together. It cannot separate out UPI's own effect, and it does not claim to.
12. Velocity cannot be measured directly. It has to be worked out from the ratio of payments or income to a stock of money, and different choices give very different answers. The turnover ratio used in Data Set 4 is an approximation, and its limits are explained where it appears.
13. Payment value is not output. A large share of UPI value is money sent between individuals. Those transfers do not buy anything newly produced, so they are not part of GDP. Any ratio of payment value to GDP has to be read with that in mind.
14. The national accounts were revised during the study period. Nominal GDP is now published on a 2022-23 base, having earlier used a 2011-12 base, so the two series do not join perfectly. This break is pointed out wherever it affects a calculation.
15. The system changes quickly. Transaction limits, fee rules and features were revised several times during the period, so the figures here are correct only up to the dates given.
16. Regional differences are hidden. National figures cover wide variation between states and districts, and UPI data is not published at district level.

3. Review of Literature

This chapter sets out what is already known. It covers how digital payments developed, the theory of the velocity of money, the research linking payments to economic activity, the policies behind UPI, and the main criticisms made of it. These are the ideas and earlier findings against which my own analysis in Chapter 4 is carried out.

3.1 Meaning and evolution of digital payments

India did not arrive at UPI in one step. Each earlier system solved one problem and left another behind.

Electronic transfers between banks came first, through RTGS and NEFT. These were reliable but built for institutions. They worked in batches or only during banking hours. The payer had to know the receiver's account number and branch code. They were meant for large payments. NPCI then introduced the Immediate Payment Service, which removed the limit of banking hours. For the first time, retail transfers could be made at any time of day.

Card networks solved a different problem, which was paying at a shop counter. But they brought costs of their own. Accepting cards needed a machine, and the shopkeeper paid a fee on every sale. Those costs shut out exactly the small traders who make up most of Indian retail. Mobile wallets then appeared and made acceptance cheaper, but they created a worse problem. Each wallet was a closed loop. Money held in one company's wallet could not reach a user of another, so the market split into networks that could not reach each other.

UPI's contribution was to treat the network itself as a public good. Instead of building a better wallet, NPCI wrote a standard that any wallet, bank or app could adopt, and made them all settle through the same system. The Bank for International Settlements calls designs like this digital public infrastructure. It means open systems that anyone can connect to and that support services used across society. The BIS treats fast payment systems as the clearest example of this, whether the operator is public or private [20].

3.2 UPI and its features

Six features set UPI apart from what came before. Together they explain the pattern of adoption described later in this chapter.

- **Interoperability.** Any app can pay any account at any member bank. By FY 2025-26 there were 703 banks on the platform, up from 44 in FY 2016-17 [2].

- No account details needed. A UPI address or a QR code replaces the account number and branch code. This removes something to remember and a common source of mistakes.
- Always available. Payment is instant and the system runs without a break, including on holidays.
- Free for the user. There is no charge to the payer. Small shopkeepers pay no Merchant Discount Rate on payments up to ₹2,000, because the Government pays them an incentive of 0.15 per cent of the value instead [14].
- Two-factor security built in. Every payment needs a PIN, and the app is tied to the customer's registered mobile number [28].
- Easy to extend. The same system now carries standing instructions, credit lines, payments abroad and UPI-linked credit cards, without changing anything the user sees.

3.3 The growth of UPI transactions in India

The full growth record appears in Data Set 1. Two points about it are worth making here, because they shape how the rest of this chapter should be read.

The first is that growth has kept going rather than spiking and flattening. A system that grows a thousand times in three years and then stops has found a small niche. A system that keeps growing for ten years has changed a habit. UPI volume rose in every single year of the period studied, including years long after the novelty had worn off.

The second is that volume has grown faster than value, year after year. This is not a statistical oddity. It is the most useful fact in the whole data set, and Data Set 2 is devoted to it. It means the typical UPI payment has been getting smaller, which tells us something about what kind of spending is moving onto the platform.

3.4 The concept of the velocity of money

Velocity is one of the oldest ideas in monetary economics and one of the most often misused. This section explains the theory carefully, because the claims made later in the project depend on getting it right.

3.4.1 The equation of exchange

Irving Fisher's equation of exchange says that the total value of payments made in an economy over a period must equal the total value of what was exchanged:

$$M \times V = P \times T$$

Here M is the money stock, V is transactions velocity, P is the average price and T is the number of transactions. This is an identity, which means it is true by the

way it is defined and not because of any assumption. Its value lies in what it forces us to see. If the money stock stays the same and the number of transactions rises, then velocity must have risen.

3.4.2 Income velocity and transactions velocity

T cannot be observed directly, so economists usually use the income form of the equation. In this version, total transactions are replaced by national income:

$$V = (P \times Y) \div M$$

This is income velocity, and it is the more cautious measure. It counts only payments made for newly produced goods and services. Transactions velocity counts every payment. That includes money moved between a person's own accounts, money repaid to a friend, and a wholesaler paying a distributor for goods that enter GDP only once, at the final sale.

The gap between the two is large, and it always runs the same way. A payment system that makes transfers easy will raise transactions velocity a great deal and income velocity much less. That is because most of the extra payments it allows are not purchases of final output. So any study that says a payment system raised "the velocity of money", without saying which velocity, should be treated with care. This project therefore reports a measure of transactions velocity in Data Set 4, and labels it clearly as one.

3.4.3 The Cambridge approach and the demand for money

The Cambridge cash-balance approach says the same thing from the other direction. It writes $M = k \times P \times Y$, where k is the share of income people choose to keep as money. Velocity is simply one divided by k . So a rise in velocity is exactly the same statement as a fall in the share of income held idle as money.

This version makes the link to payment technology easy to see. Keynes pointed out that people hold money for transactions because income and spending do not arrive at the same time. They need a balance to bridge that gap. A system that lets them transfer money instantly, at any hour, from an account that earns interest makes the bridge smaller. So balances fall compared with spending, k falls, and V rises. This is how UPI could plausibly affect velocity, and it is what this project tests.

3.4.4 What velocity does not tell us

One warning is needed. A rise in velocity is not by itself a sign of a healthy economy. Velocity rises sharply during hyperinflation, when holding money is punished and people spend at once. It falls in uncertain times, when people save for safety. Faster circulation is only a good sign when it comes from lower costs of

paying, and not from people fleeing money itself. This difference matters for how Data Set 4 is read.

3.5 Digital payments and the circulation of money

The Reserve Bank treats the link between digital payments and the demand for cash as a policy matter, not just an academic one. Payments Vision 2025 lists reducing cash in circulation, as a share of GDP, among its goals. It notes that India's ratio is high compared with other countries, and that the demand for cash grows with both output and inflation [26].

Digital payments do not reduce the demand for cash by making people poorer. They change the form in which wealth is held. Money kept in a bank account, and spendable instantly through a phone, does the job cash used to do. The difference is that it stays inside the banking system, where it can be lent out. The Reserve Bank tracks this shift through its Digital Payments Index, which uses March 2018 as a base of 100. It stood at 465.33 in September 2024 and 493.22 in March 2025 [6].

Evidence on household behaviour points the same way. The Department of Financial Services study of February 2026 found that users of UPI and RuPay reported a clear fall in both cash use and ATM withdrawals. Of those users, 74 per cent said speed of payment was the main advantage [15].

3.6 Digital payments and economic activity

Research on whether fast payment systems affect real activity is still developing. It is also more careful than most popular writing on the subject.

The Bank for International Settlements studied 86,163 apps across 95 countries between 2012 and 2022. It found that launching a fast payment system increases the use of digital finance apps, and that the effect is largest in lower-income countries. The effect was strongest for apps built by new fintech and big-tech firms rather than by established banks. It was also larger where the central bank took an active role, where membership was open to all, and where settlement happened in real time [20]. India's system has all three of those features.

The International Monetary Fund studied interoperability using data on every UPI transaction. It found that digital payments later grew fastest in those districts where the payment market had been most divided beforehand, and where joining it together therefore helped most [18]. This comes closer to showing cause and effect than most work in the field, because it compares districts that received different sized changes rather than simply comparing before with after.

Evidence from similar economies points the same way. A study of thirty-three Indonesian provinces found that digital payments had a significant effect on regional income and consumption, both before and after a structural break linked to COVID-19. The effect was larger after the break [29]. Indonesia is not India, so this finding cannot simply be transferred. It is quoted here to show that the relationship can be detected in this kind of data, not as a measurement of what happened in India.

For India itself, the most useful evidence is about behaviour rather than national figures. One study surveyed 276 people and followed up with 20 interviews. About 75 per cent said they spent more because of UPI. Many explained that digital money felt less real, so they felt less reluctant to part with it. In the same study, 95.2 per cent found UPI convenient and 91.5 per cent were satisfied with it [25]. The sample is small and the answers are self-reported, so the results cannot be applied to the whole country. But the study shows something national figures cannot see, and it carries a warning as well as a finding. A system that makes spending easier also makes overspending easier.

The evidence from shopkeepers is stronger. The Department of Financial Services study found that 94 per cent of small merchants had adopted UPI. About 72 per cent were satisfied with digital payments, giving faster transactions and better record-keeping as their reasons. Most importantly for this project, 57 per cent said their sales had risen after they adopted digital payment [15]. These are reported figures, not audited accounts, and a shopkeeper may credit a payment method for a gain that had other causes. Even so, a majority reporting higher sales is the most direct evidence available that the effect reaches real activity, and not just the way payments are made.

3.7 Financial inclusion and the digital economy

Financial inclusion in India was built in layers, and the payment layer is what made the others useful. A Jan Dhan account that is opened and never used is just an entry in a register. The same account, connected to a payment system that a vegetable seller will accept, becomes a working financial relationship.

The Reserve Bank's Financial Inclusion Index records this change. It rose from 64.2 in March 2024 to 67.0 in March 2025, and all three of its parts improved: access, usage and quality [7]. The rise in usage is the one that matters most. Access had already been largely achieved through the Jan Dhan scheme, which had reached over 55.98 crore people by August 2025, more than 55 per cent of them women [7].

The World Bank's Global Findex 2025 surveyed about 145,000 adults in 141 countries and provides the international standard for measuring this. It records one Indian finding that is directly relevant. The share of Indian women holding inactive accounts fell from a third in 2021 to 18 per cent by 2024 [21]. Inactive accounts are exactly the problem that a usable payment system solves.

Beyond payments, the record of transactions has itself become valuable. The Reserve Bank built the Unified Lending Interface on this foundation. It gives approved lenders easy access to verified information about borrowers, which turns a payment history into proof of creditworthiness [20]. For a small trader who has never had audited accounts, this is how accepting digital payments can become access to a loan.

3.8 Government initiatives promoting digital payments

Adoption on this scale did not happen on its own. Four kinds of government action shaped it.

- **Pricing.** The Government has kept UPI free for users and has paid the system for the income it gives up, rather than letting shopkeepers be charged. Several schemes have paid banks a percentage of the transaction value in place of a Merchant Discount Rate, including a ₹1,500 crore scheme for small merchant payments [14], [15].
- **Infrastructure.** The Payments Infrastructure Development Fund pays part of the cost of putting acceptance infrastructure into smaller towns. By 31 October 2025 about 5.45 crore digital payment points had been set up through the Fund in tier-3 to tier-6 centres. By FY 2024-25, some 56.86 crore QR codes had been given to about 6.5 crore merchants [14].
- **Identity and accounts.** The Jan Dhan, Aadhaar and Mobile combination provided what was needed first: an account to receive money, an identity to prove who the holder was, and a phone to pay from.
- **Going international.** Through NPCI International Payments Limited, UPI payments and money transfers have been extended to eleven foreign countries. Acceptance began in Cambodia in June 2026, and a money transfer link with the Maldives' instant payment system started in July 2026 [17].

The cost of this policy should be stated honestly. Free payments are not costless payments. The cost has simply moved from the shopkeeper to the Government. Whether that is worth it depends on whether the gains in recorded activity, inclusion and tax collection are larger than the amount spent. This project raises that question in Chapter 5 but cannot answer it.

3.9 Challenges faced by digital payment systems

Four problems come up again and again in the research and in policy documents. An honest assessment has to give them proper weight.

Cyber fraud and tricking users.

Cybersecurity incidents in India rose from 10.29 lakh in 2022 to 22.68 lakh in 2024 [28]. The technology itself is not the main weak point. Tying the app to one device, and requiring a PIN, make it hard to break in. The weak point is the user, who can be tricked into approving a payment. NPCI runs a fraud-monitoring system that uses machine learning to raise alerts and block suspicious payments. The Citizen Financial Cyber Fraud Reporting and Management System has saved more than ₹1,200 crore across more than 4.7 lakh complaints [28]. A system used by over 55 crore people attracts attacks in proportion to its size.

Dependence on infrastructure.

A digital payment needs electricity, a working phone and a network signal, all at the same moment. Cash needs none of these. So as UPI replaces cash, the risk becomes concentrated. A breakdown that would once have been an inconvenience now stops trade.

Unequal access.

Data Set 8 measures the digital divide. In August 2025 rural areas had 46.73 internet subscribers per 100 people, against 113.83 in urban areas [22]. A payment system that can only be reached through the internet cannot include more people than the internet itself reaches.

Risk to spending habits.

The finding that about 75 per cent of surveyed users spent more, because digital money felt less real [25], is a genuine cost. It matters most for low-income households, who have little room to absorb overspending. The same ease that raises velocity also removes a restraint that cash used to provide by accident.

3.10 Case studies

3.10.1 Urban and rural India

The rural story is one of fast catching up from a low start, achieved by deliberately paying for acceptance infrastructure rather than waiting for the market. The Payments Infrastructure Development Fund exists for exactly this reason. Putting a QR code in a tier-6 town is not profitable at the small volumes available there at first. By October 2025 the Fund had set up 5.45 crore payment points in tier-3 to

tier-6 centres [14]. The gap that remains is not mainly about acceptance, though. It is about connectivity, and Data Set 8 measures it.

3.10.2 Small businesses and consumers

For the small shopkeeper, UPI solved a problem that cards never could. A card machine costs money, and a fee on every sale eats into a thin margin. A printed QR code costs nothing, and below ₹2,000 there is no fee at all [14]. The result is that almost everyone accepts it. Of small merchants surveyed, 94 per cent had adopted UPI and 57 per cent said their sales rose afterwards [15]. Street vendors under the PM SVANidhi scheme have been brought into the same system, with cashback on digital payments and access to UPI-linked RuPay credit cards that turn a payment record into a line of credit [14].

3.10.3 India in international comparison

India stands out in two ways. The first is size. UPI handled close to 49 per cent of all real-time payments made in the world in 2024, and the IMF calls it the largest retail fast payment system by volume [3], [18]. The second is pricing. Most similar systems abroad charge shopkeepers something. The BIS notes that in India there is no charge to the user for UPI, and a fee applies only to merchant payments made through prepaid instruments like wallets [20]. India chose to treat retail payment as a public service, and the adoption described in this chapter is partly a result of that choice.

4. Analysis of Data

This chapter contains the original analysis of the project. Eight data sets are examined. Each one follows the same four-part structure used throughout: the table or graph, an explanation of what the data show, an analysis of what they mean, and an interpretation saying what can and cannot be concluded. Ten figures and eight tables support the discussion, and each data set ends with one sentence summing up its result.

The first four data sets look at growth, composition and velocity. Data Sets 5 and 6 test the two big shocks of the period. Data Sets 7 and 8 look at financial inclusion and at its limits. Together they test the three hypotheses given in Section 2.2.

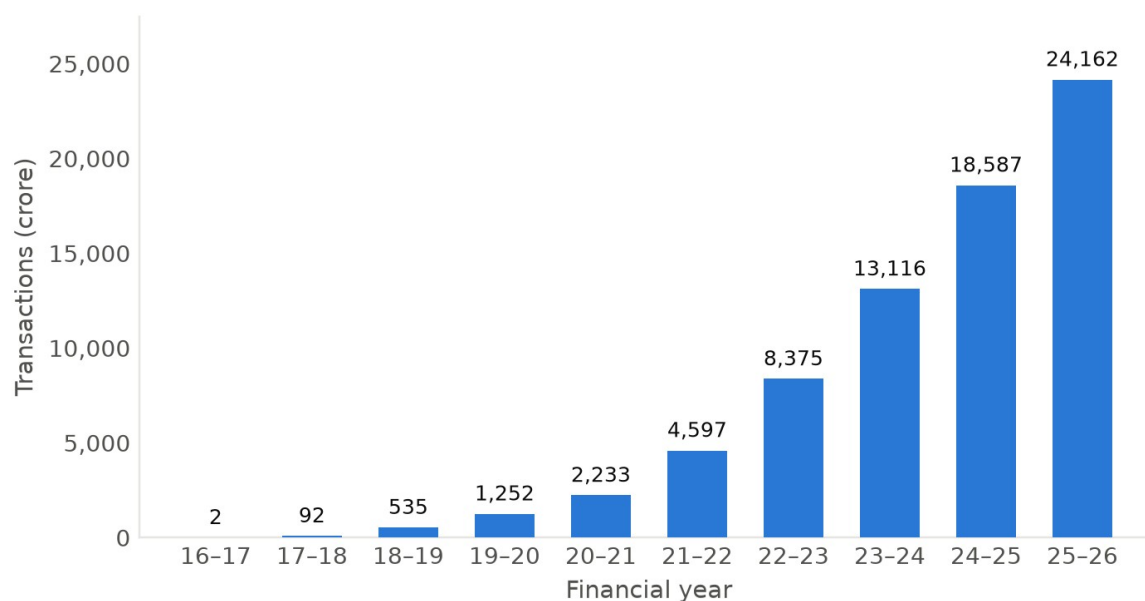
4.1 Data Set 1 · The growth of UPI transactions

Financial year	Volume (crore)	Value (₹ lakh crore)	Volume growth	Value growth
2016-17	2	0.07	—	—
2017-18	92	1.10	4,500%	1,471%
2018-19	535	8.77	482%	697%
2019-20	1,252	21.32	134%	143%
2020-21	2,233	41.04	78%	92%
2021-22	4,597	84.16	106%	105%
2022-23	8,375	139.00	82%	65%
2023-24	13,116	199.90	57%	44%
2024-25	18,587	261.00	42%	31%
2025-26	24,162	314.00	30%	20%

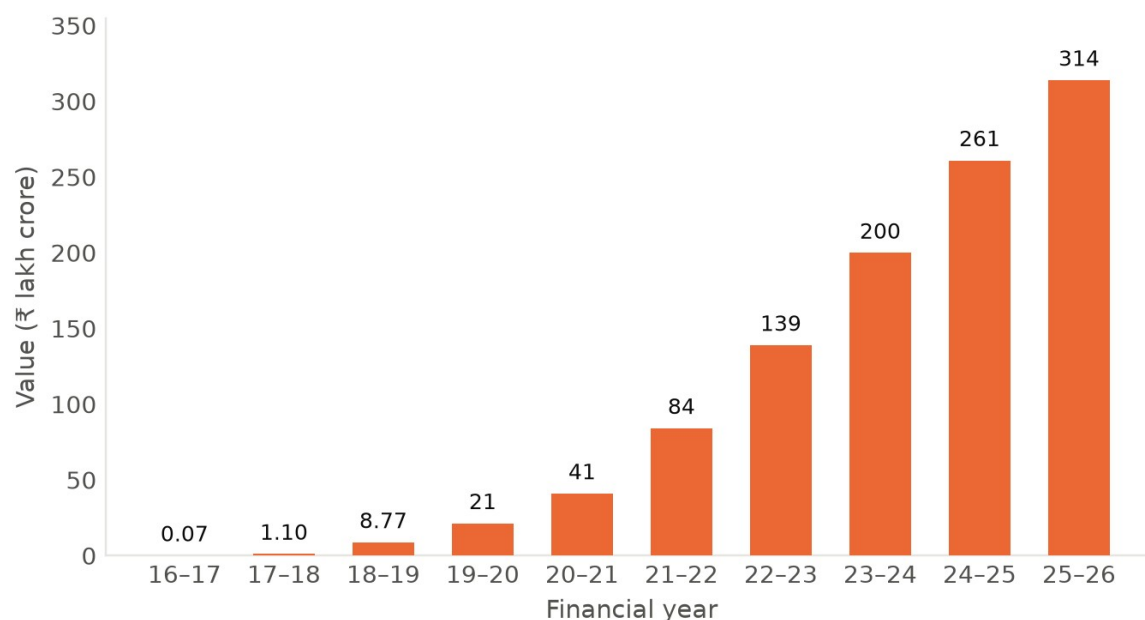
Table 1. Sources: NPCI UPI product statistics [1], with year-wise figures as reported in Press Information Bureau releases [2], [10], [13], [16]. Growth rates are computed by the author from the volume and value columns. Figures are rounded as published.

Figure 1 UPI transaction volume, FY 2016-17 to FY 2025-26

Annual number of UPI transactions, in crore

*Figure 1. UPI transaction volume, FY 2016-17 to FY 2025-26.***Figure 2 UPI transaction value, FY 2016-17 to FY 2025-26**

Annual value of UPI transactions, in ₹ lakh crore

*Figure 2. UPI transaction value, FY 2016-17 to FY 2025-26.***Explanation of the data**

The table shows the number and value of UPI transactions in each financial year since the platform started. It also gives the year-on-year growth rates, which I

calculated from those two columns. Volume rose from 2 crore transactions to 24,162 crore. Value rose from ₹0.07 lakh crore to about ₹314 lakh crore. Both changes happened over ten years [2].

Analysis

Three patterns stand out. The first is that growth rates are slowing in percentage terms, from 4,500 per cent down to 30 per cent. But the actual increase each year keeps getting bigger. The rise in FY 2025-26 alone was about 5,575 crore transactions, which is more than the entire volume of FY 2021-22. This is just what happens as a base grows larger. It should not be read as a real slowdown.

The second is that the growth never flattens out. Ten years of rising volume, through a pandemic and long after the novelty wore off, points to a lasting change in how people pay rather than a passing fashion.

The third pattern matters most. Volume has grown faster than value in every year since FY 2021-22. In FY 2025-26 volume grew 30 per cent while value grew 20 per cent. The average size of a transaction is value divided by volume, so if volume grows faster, the average payment must be getting smaller. Data Set 2 looks at this directly.

Interpretation

UPI has gone from an experiment to the normal way India pays in shops, and it now handles most retail digital transactions. The gap between volume growth and value growth is the first sign of something more. It suggests UPI is not simply taking over payments that already existed in another form. It is reaching very small payments that were previously made in cash, or not made at all. Hypothesis H1 is supported so far as a description. Its effect on velocity is tested in Data Set 4.

In one sentence: UPI grew from 2 crore transactions to 24,162 crore in ten years, and because volume grew faster than value in every year from FY 2021-22, the typical payment was getting smaller even as the total kept rising.

4.2 Data Set 2 · The average value of a single UPI transaction

Financial year	Value (₹ lakh crore)	Volume (crore)	Average per transaction (₹)
2017-18	1.10	92	1,196
2018-19	8.77	535	1,639

Financial year	Value (₹ lakh crore)	Volume (crore)	Average per transaction (₹)
2019-20	21.32	1,252	1,703
2020-21	41.04	2,233	1,838
2021-22	84.16	4,597	1,831
2022-23	139.00	8,375	1,660
2023-24	199.90	13,116	1,524
2024-25	261.00	18,587	1,404
2025-26	314.00	24,162	1,300

Table 2. The average value per transaction is derived by the author as transaction value divided by transaction volume, using the figures in Table 1. FY 2016-17 is omitted because the base is too small for the ratio to be meaningful. Underlying data: [1], [2].

Figure 3 Average value of a single UPI transaction

Transaction value divided by transaction volume; FY 2016-17 excluded (negligible base)

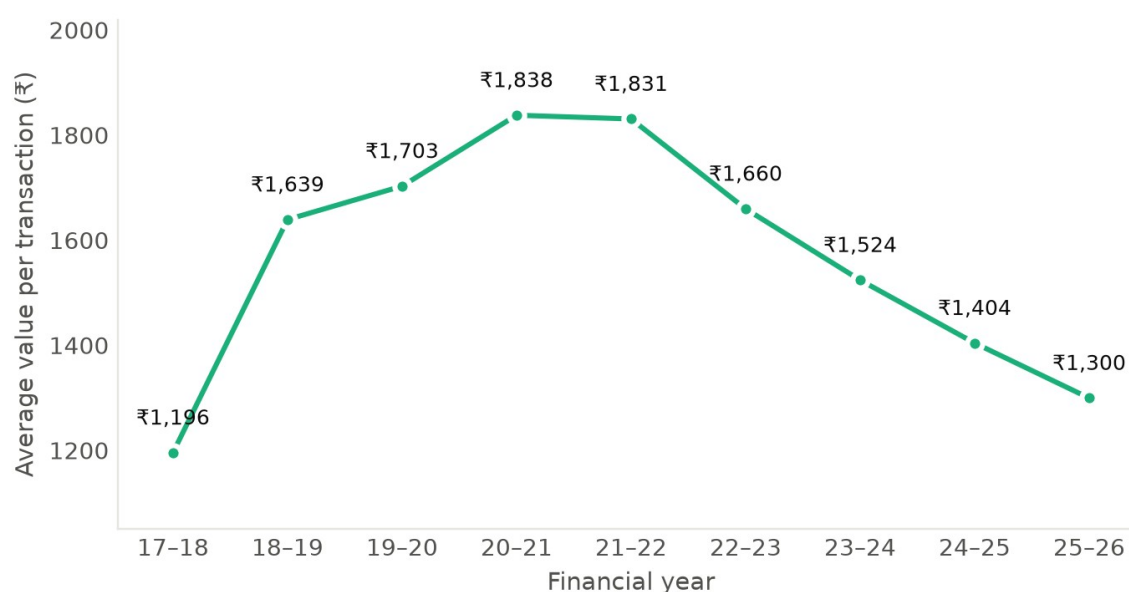


Figure 3. Average value of a single UPI transaction, FY 2017-18 to FY 2025-26.

Explanation of the data

This figure is not published anywhere. I worked it out by dividing each year's transaction value by its volume. It answers a simple question: how big is a typical UPI payment, and is that changing? The answer rises from ₹1,196 in FY 2017-18 to a peak of ₹1,838 in FY 2020-21. It then falls steadily to ₹1,300 in FY 2025-26, a drop of about 29 per cent from the peak.

Analysis

The shape of this curve tells the story of UPI's adoption in two clear phases.

In the first phase, up to FY 2020-21, the average payment grew. This is what you would expect when the early users are better off, live in cities and are comfortable with technology. The payments moving onto UPI at that stage were large ones, such as rent, fees and transfers between family members, which used to be made by cheque or bank transfer.

In the second phase the average falls, every year, for five years. Total value kept rising throughout, so this cannot mean people were spending less. It can only mean that the new payments being added were much smaller than the existing average, and that there were enough of them to pull the average down. The composition data confirms this. Person-to-merchant payments are now 63 per cent of all UPI volume, and 86 per cent of those are below ₹500 [2].

Inflation works against this finding rather than causing it. Prices rose over the period, so the same basket of goods would show a rising average payment. The fact that the average fell anyway means the shift towards small payments is even larger in real terms than it looks.

Interpretation

This is the most important finding in the project, because it separates the two possible explanations of UPI's growth. If UPI had only digitised payments that already existed, the average payment size would have stayed roughly flat while volume rose. Instead it fell by nearly a third while volume multiplied thirteen times. That is the mark of a payment system reaching transactions it never served before: the tea, the vegetables, the auto fare.

This matters directly for velocity. Fisher's equation links the money stock to the number of transactions it supports. A system that adds a very large number of very small payments raises T a lot while adding relatively little to the total value settled. That is exactly what it means for each rupee to be used more often. Hypothesis H2 is supported.

In one sentence: The average UPI payment fell from ₹1,838 to ₹1,300 while volume rose more than ten-fold, which is the signature of a system adding small everyday payments rather than merely digitising ones that already existed.

4.3 Data Set 3 · UPI within India's wider digital payment system

Indicator	Earlier period	Later period
Total digital payment volume (crore)	2,071 (FY 2017-18)	18,737 (FY 2023-24)

Indicator	Earlier period	Later period
Total digital payment value (₹ lakh crore)	1,962 (FY 2017-18)	3,659 (FY 2023-24)
Volume CAGR over the period	—	44%
Value CAGR over the period	—	11%
UPI share of retail digital payment volume	—	81% (FY 2024-25)
Digital transactions, FY 2019-20 to FY 2024-25	—	over 65,000 crore, worth over ₹12,000 lakh crore

Table 3. Sources: Press Information Bureau releases [10], [11], [12], [14].

Figure 5 UPI's share of retail digital payment volume, FY 2024-25

Share of the total number of retail digital payment transactions in India

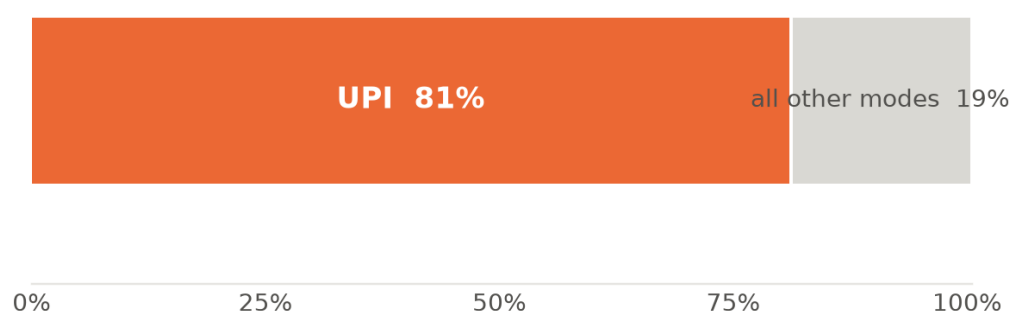


Figure 5. UPI's share of retail digital payment volume, FY 2024-25.

Explanation of the data

This data set places UPI within all digital payments in India, including cards, wallets, NEFT, IMPS and RTGS. Total digital payment volume rose from 2,071 crore transactions in FY 2017-18 to 18,737 crore in FY 2023-24, a compound growth rate of 44 per cent a year. Value grew from ₹1,962 lakh crore to ₹3,659 lakh crore, a compound rate of 11 per cent [10], [11]. UPI alone made up 81 per cent of retail digital payment volume in FY 2024-25 [14].

Analysis

The difference between those two growth rates, 44 per cent for volume against 11 per cent for value, repeats across the whole payment system the pattern found in UPI alone. India's digital payments have grown mainly through more payments being made, not through bigger ones.

The value figures need care. Total digital payment value of ₹3,659 lakh crore is far larger than India's GDP. That is because it includes RTGS, which carries very

large transfers between banks and big companies. There are not many of these, but each is enormous. This shows clearly why payment value cannot be treated as economic output, and why the ratio in Data Set 4 has to be called a transactions measure rather than income velocity.

There is no contradiction in UPI carrying 81 per cent of retail transactions by number while total value is dominated by large wholesale systems. It simply describes what UPI is: the system for frequent, small, everyday payments.

Interpretation

UPI is not one choice among several in Indian retail payments. It is the system, with everything else sharing the remaining fifth. For this study, that means changes in retail payment behaviour over the period can fairly be linked to UPI, because no alternative is large enough to drive the national figures.

In one sentence: UPI carries 81 per cent of India's retail digital transactions, so changes in retail payment behaviour during this period can reasonably be attributed to it.

4.4 Data Set 4 · UPI turnover relative to nominal GDP

Financial year	UPI value (₹ lakh crore)	Nominal GDP (₹ lakh crore)	UPI ÷ GDP
2022-23	139.00	269.50	0.52×
2023-24	199.90	295.36	0.68×
2024-25	261.00	318.07	0.82×
2025-26	314.00	346.36	0.91×

Table 4. UPI values from [1], [2]. Nominal GDP at current prices from MoSPI provisional estimates [8], [9]. The ratio is computed by the author. Note a discontinuity: figures for FY 2022-23 and FY 2023-24 are on the 2011-12 base series, while FY 2024-25 and FY 2025-26 are on the revised 2022-23 base series. The break is small relative to the trend but the series is not perfectly continuous.

Figure 4 UPI turnover relative to nominal GDP

A transactions-velocity proxy: rupees moved over UPI per rupee of national output

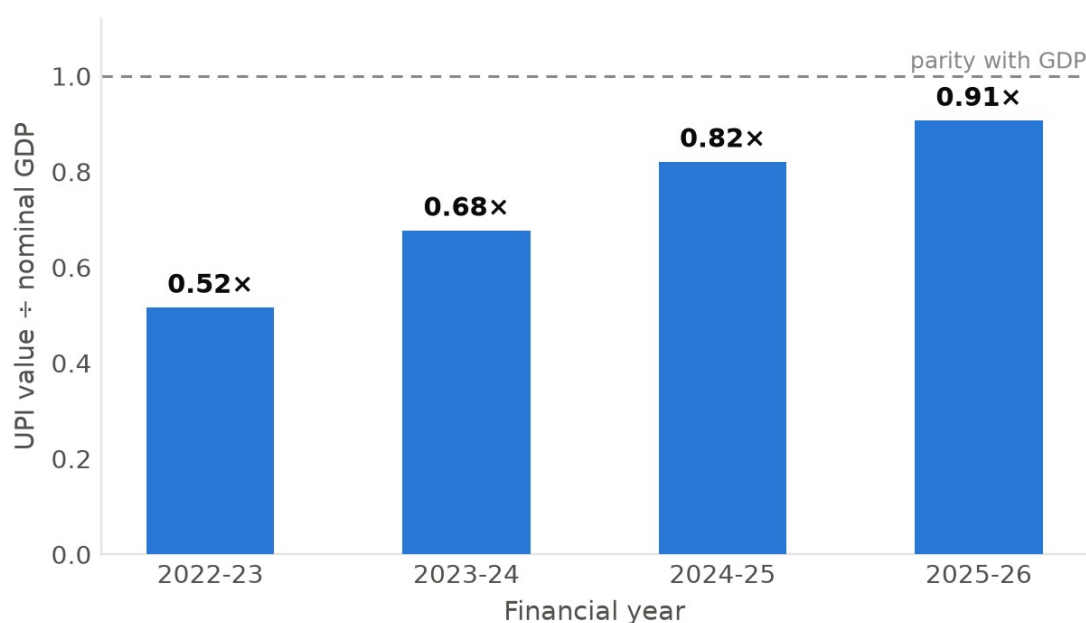


Figure 4. UPI turnover relative to nominal GDP, FY 2022-23 to FY 2025-26.

Explanation of the data

This data set builds the closest thing to a velocity measure that public data allows. It shows the total value settled over UPI in a year as a multiple of nominal GDP in the same year. The ratio rose from 0.52 in FY 2022-23 to 0.91 in FY 2025-26. By the latest year, the value passing through UPI alone came close to the entire annual output of the Indian economy.

Analysis

It is important to say exactly what this ratio does and does not measure, because it would be easy to claim too much from it.

It is a measure of transactions turnover, not income velocity. The top of the ratio counts every payment, including transfers between people that do not buy anything newly produced, and payments between businesses that GDP counts only once at the final sale. The bottom counts final output only. The two are not measuring the same thing, so the number should not be read as "each rupee now circulates 0.91 times".

What the ratio does measure reliably is the direction and speed of change. The mismatch between the top and the bottom stays roughly the same from year to year, so the comparison between years is still fair. On that basis the ratio rose by about 75 per cent in three years. The value settled through this one retail system grew far faster than national output. Payment activity per unit of output has

clearly increased, which is what the Cambridge version predicts: as paying gets cheaper, people hold less money against a given amount of spending, so turnover rises.

Two cautions are needed. First, part of the rise is simply cash payments moving onto UPI. That raises measured turnover without any change in behaviour, because cash payments were never counted in the first place. Second, as Section 3.4.4 explained, rising velocity is only a good sign when it comes from cheaper transactions rather than from people fleeing money. Here the cause is clearly the first. India had no monetary crisis in this period, and the rise matches the documented fall in the cost and difficulty of paying.

Interpretation

Hypothesis H1 is supported, with one qualification. What has been shown is a rise in measured transactions turnover, not in income velocity. Part of that rise is cash payments becoming visible for the first time rather than new activity. Both effects are real and both matter, but they are not the same thing, and the published data cannot separate them. This is the honest limit of what a study based on national figures can establish.

In one sentence: The value settled over UPI rose from 0.52 to 0.91 times India's GDP in three years — the level of that ratio is not meaningful, but its rapid rise is, and it points to a higher transactions velocity of money.

4.5 Data Set 5 · Digital payments around demonetisation

Indicator	Value
Value of specified bank notes demonetised (Nov 2016)	₹15.4 trillion
Share of total value of notes in circulation	86.9%
Share of demonetised notes returned to the RBI	approximately 98.96%
Monthly digital transactions, October 2016	71.27 crore
Monthly digital transactions, May 2017	111.45 crore
Increase over the period	56%

Table 5. Sources: RBI, *Macroeconomic Impact of Demonetisation* [23]; PIB [24].

Figure 6 Digital payments around demonetisation

Monthly digital transaction volume, October 2016 vs May 2017

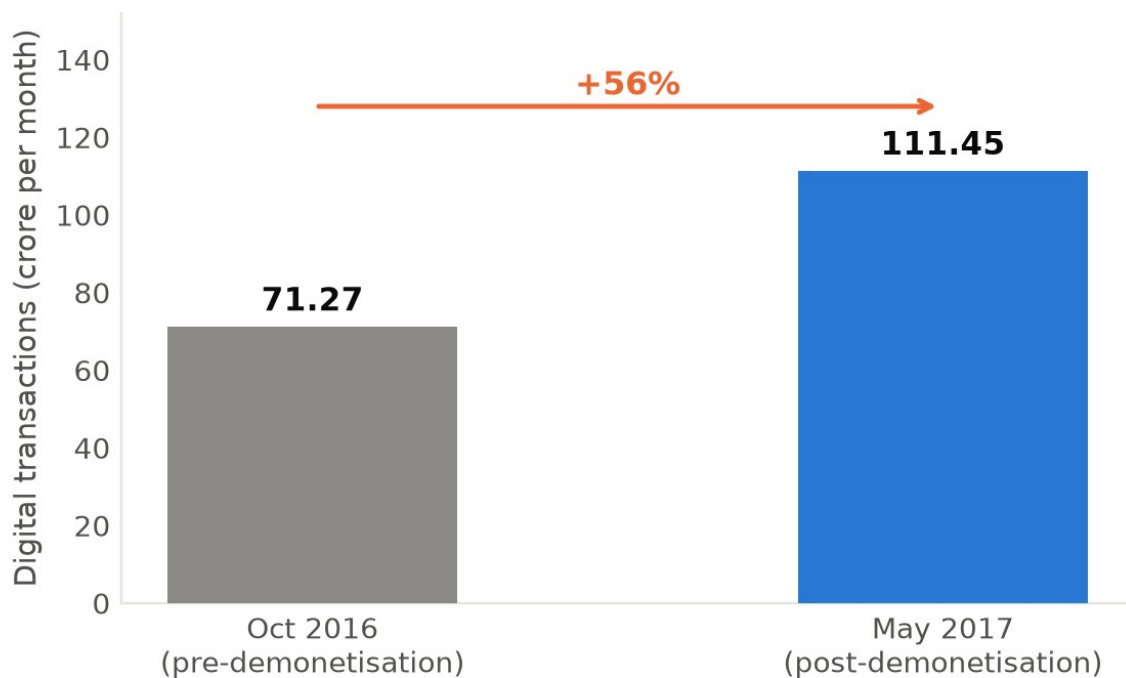


Figure 6. Monthly digital transaction volume before and after demonetisation.

Explanation of the data

On 9 November 2016, ₹500 and ₹1,000 notes stopped being legal tender. They were worth ₹15.4 trillion, which was 86.9 per cent of the value of all notes then in circulation [23]. Monthly digital transactions rose from 71.27 crore in October 2016 to 111.45 crore by May 2017, an increase of 56 per cent [24].

Analysis

Demonetisation worked like an unusually severe natural experiment. The main way people paid was withdrawn almost overnight, and an alternative had been launched only weeks before. The 56 per cent rise in digital transactions over seven months is how people responded.

That response should not be overstated. About 98.96 per cent of the withdrawn notes came back to the banking system [23]. This means the currency was replaced rather than permanently removed, and cash use recovered a great deal in the years that followed. Demonetisation did not by itself turn India into a digital economy.

What it did achieve is better described as a forced trial. Many people who would not otherwise have installed a payment app did so because they had to. Many shopkeepers who would not otherwise have accepted digital payment learned

how. Some of that behaviour stayed once the pressure was removed. The lasting effect was on familiarity and infrastructure rather than on how people actually paid.

Interpretation

Demonetisation speeded up UPI's adoption at an early and important moment, but it did not cause the growth that followed. That growth continued for years after cash supply returned to normal, and at rates far above anything seen in 2016-17. The episode is better understood as removing a barrier, which was the effort of learning something new, rather than changing the economics of paying.

In one sentence: Demonetisation produced a 56 per cent jump in digital payments, but about 99 per cent of the currency came back, so it accelerated UPI's adoption without causing its sustained growth.

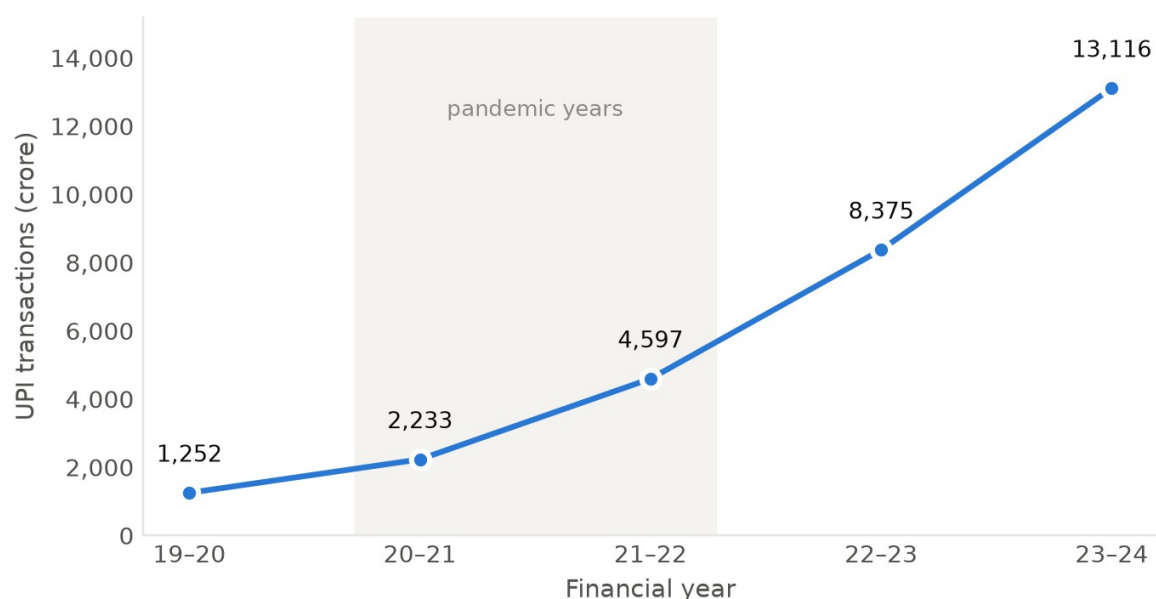
4.6 Data Set 6 · UPI growth before, during and after COVID-19

Financial year	Volume (crore)	Growth	Period
2019-20	1,252	134%	Pre-pandemic
2020-21	2,233	78%	Pandemic
2021-22	4,597	106%	Pandemic
2022-23	8,375	82%	Post-pandemic
2023-24	13,116	57%	Post-pandemic

Table 6. Volumes from [1], [2], [10], [13]. Growth rates computed by the author.

Figure 7 UPI volume before, during and after COVID-19

FY 2019-20 to FY 2023-24; shaded band marks FY 2020-21 and FY 2021-22

*Figure 7. UPI transaction volume before, during and after the COVID-19 pandemic.***Explanation of the data**

This table separates out the pandemic years. UPI volume grew 78 per cent in FY 2020-21, the year of the strictest restrictions, and then 106 per cent in FY 2021-22, which is a higher rate than the year before. Volume more than tripled between FY 2019-20 and FY 2021-22.

Analysis

The pattern looks surprising at first. India's real output shrank in FY 2020-21, and you would expect a payment system to shrink along with the economy it serves. Instead UPI volume grew 78 per cent. Two forces explain this.

The first is substitution. With movement restricted and physical contact discouraged, payments that would have been made in cash were made digitally instead. This moves payments between methods without increasing how many there are.

The second is adoption, and it matters more. Shopkeepers who had refused digital payment accepted it because customers insisted. Customers who had avoided it learned because there was no other option. This permanently widened the number of people who could use the system.

The proof that adoption mattered more than substitution lies in what happened next. If the pandemic rise had been pure substitution, growth would have reversed once normal trade resumed. Instead FY 2021-22 grew faster than FY

2020-21, and growth continued at 82 per cent and 57 per cent in the two following years, from a much larger base. The change did not unwind.

Interpretation

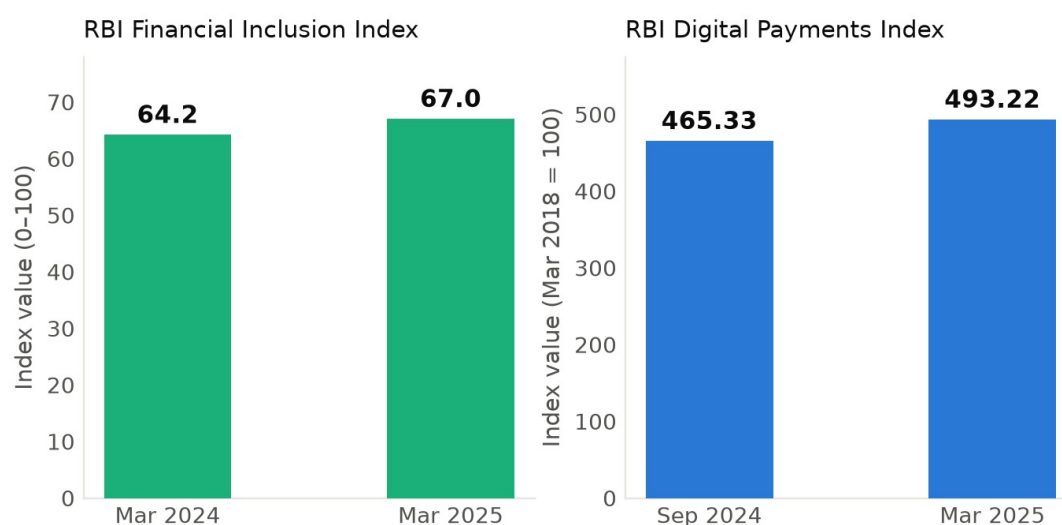
COVID-19 acted as a second shock, and a more lasting one than demonetisation. The contrast between the two is useful. Demonetisation removed the alternative for a few months, and adoption partly reversed afterwards. The pandemic lasted long enough for digital payment to become a habit. Habits, unlike forced compliance, survive the removal of the pressure that created them.

In one sentence: UPI grew faster after the pandemic (106 per cent) than during its worst year (78 per cent), which shows the change had become a habit rather than reverting once cash was available again.

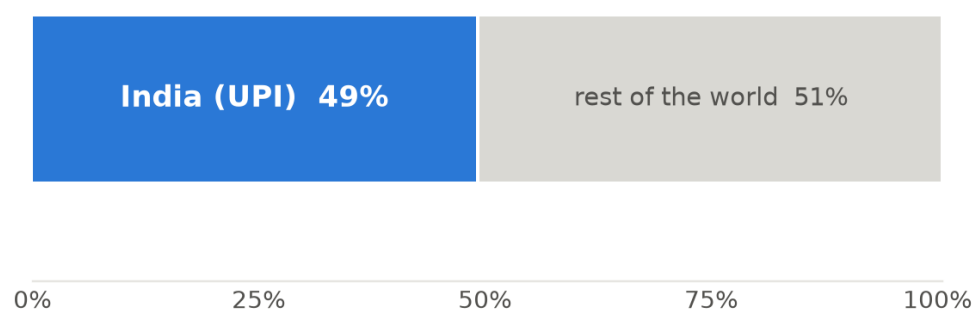
4.7 Data Set 7 · Financial inclusion, digitalisation and global standing

Indicator	Earlier	Later
RBI Financial Inclusion Index	64.2 (Mar 2024)	67.0 (Mar 2025)
RBI Digital Payments Index (Mar 2018 = 100)	465.33 (Sep 2024)	493.22 (Mar 2025)
PMJDY beneficiaries	—	55.98 crore (Aug 2025)
Share of PMJDY accounts held by women	—	over 55%
UPI users onboarded	—	55.49 crore (Jun 2026)
Merchants accepting UPI	—	approx. 6.5 crore (FY 2024-25)
QR codes deployed	—	56.86 crore (FY 2024-25)
India's share of global real-time payment volume	—	approx. 49% (2024)

Table 7. Sources: RBI and PIB releases [3], [6], [7], [14], [16]; IMF [18].

Figure 8 Financial inclusion and payment digitalisation, RBI indices*Figure 8. RBI Financial Inclusion Index and RBI Digital Payments Index.***Figure 10 India's share of global real-time payment volume, 2024**

As recognised by the IMF, June 2025

*Figure 10. India's share of global real-time payment volume, 2024.***Explanation of the data**

This data set brings together the main indicators of financial inclusion and digitalisation. Both RBI indices rose over their latest measurement periods, and the Financial Inclusion Index improved in all three of its parts: access, usage and quality [7]. The last row records that India handled about 49 per cent of all real-time payment transactions in the world in 2024 [3].

Analysis

The rise in the usage part matters more than the headline number. Access had already been largely achieved through the Jan Dhan scheme. The real problem was accounts lying unused, and the World Bank records that the share of Indian women with inactive accounts fell from a third in 2021 to 18 per cent by 2024

[21]. A payment system that a small vendor will accept is what turns an account from a registration into something useful.

The international comparison needs to be kept in proportion. India's 49 per cent share of world real-time payments partly reflects its population, and a share of volume says nothing about quality. What makes it notable is the policy difference the BIS points out. Most fast payment systems abroad charge shopkeepers, and India's does not [20]. India chose to treat retail payment as a public service, and the adoption pattern is partly a result of that choice.

The closeness of 55.98 crore Jan Dhan accounts and 55.49 crore UPI users is worth noticing. These are different groups measured on different dates, so they cannot simply be set against each other. But their similar size suggests that owning an account and being able to pay digitally now reach a comparable share of adults.

Interpretation

The gains in financial inclusion are real and can be measured. The chain running from accounts, through usage, to activity being recorded is visible in the indices rather than just claimed. India's international standing rests on a deliberate decision about pricing, not on size alone.

In one sentence: The RBI's Financial Inclusion Index rose from 64.2 to 67.0 with usage improving fastest — the sub-index that matters, because access had already been achieved and dormancy was the real problem.

4.8 Data Set 8 · Internet penetration and the rural-urban divide

Indicator	Urban	Rural
Internet subscribers, Aug 2025 (million)	579.46	423.39
Internet subscribers per 100 people, Aug 2025	113.83	46.73
Tele-density, Dec 2025	148.92%	59.63%
Telecom subscribers, Dec 2025 (million)	762.44	543.70

Table 8. Source: TRAI telecom subscription data and performance indicator reports [22]. Total broadband subscribers stood at 999.81 million as at 31 October 2025, and total telecom subscribers at 1,306.14 million as at 31 December 2025, giving an overall tele-density of 91.74 per cent.

Figure 9 The rural-urban digital divide

Urban internet penetration is roughly 2.4 times the rural rate

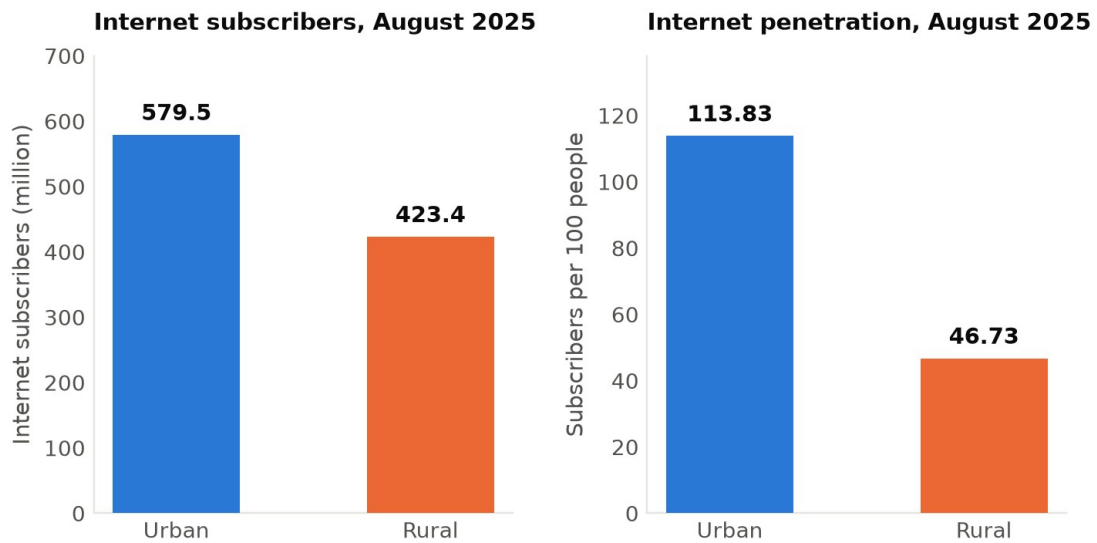


Figure 9. Internet subscribers and internet penetration, rural and urban India.

Explanation of the data

UPI needs a connected device, so internet access sets the limit on how inclusive it can be. India had 999.81 million broadband subscribers on 31 October 2025 [22]. In urban areas there were 113.83 internet subscribers per 100 people. In rural areas there were 46.73, which is a ratio of roughly 2.4 to 1.

Analysis

The urban figure is above 100 per 100 people because it counts subscriptions, not people, and many urban users have more than one. This is a reminder to read the rural figure as an upper limit. If some rural users also hold more than one subscription, then fewer than 46.73 per cent of rural people are actually online.

In absolute numbers the gap is smaller than the penetration figures suggest: 579.46 million urban subscribers against 423.39 million rural ones. That is because rural India has more people. The two panels of Figure 9 therefore tell different halves of the same story. Rural India is a large and growing market in total, but a much smaller share of rural people is reached.

This is a limit of connectivity, not of acceptance. The Payments Infrastructure Development Fund has set up about 5.45 crore digital payment points in tier-3 to tier-6 centres [14], so the problem is not that rural shops lack QR codes. The problem is that a large share of rural people cannot reliably reach the network those codes depend on.

Interpretation

Hypothesis H3 is supported. The benefits of UPI described throughout this chapter are shared unequally, and the inequality is built into the infrastructure rather than being a matter of choice. It follows the spread of connectivity. So any policy meant to extend the gains of digital payment further into rural India has to deal with the telecom problem, because payment measures alone cannot solve it.

In one sentence: Rural internet penetration is 46.73 per 100 people against an urban 113.83, so what limits UPI's reach in rural India is connectivity, not merchant acceptance.

5. Findings and Suggestions

5.1 Findings

Seven findings come out of the analysis in Chapter 4. Each one is given with the evidence behind it and with the qualification that evidence requires.

Finding 1. UPI has become the normal way India pays in shops.

Volume rose from 2 crore transactions in FY 2016-17 to 24,162 crore in FY 2025-26. UPI made up 81 per cent of retail digital payment transactions in FY 2024-25 [2], [14]. Growth happened in every year of the period, and the actual increase each year keeps rising even as the percentage growth slows.

Finding 2. The growth is made up of small payments, and increasingly so.

The average value of a UPI payment fell from ₹1,838 in FY 2020-21 to ₹1,300 in FY 2025-26, a fall of about 29 per cent. Over the same years volume rose more than ten times (Data Set 2). Person-to-merchant payments are now 63 per cent of volume, and 86 per cent of those are below ₹500 [2]. Since prices rose over the period, the shift towards small payments is even larger in real terms than the figures show.

Finding 3. The value of payments settled for each rupee of output has risen sharply.

UPI value as a multiple of nominal GDP rose from 0.52 in FY 2022-23 to 0.91 in FY 2025-26 (Data Set 4). This fits the Cambridge prediction that when paying becomes cheaper, people hold less money against a given amount of spending. It is a transactions measure, not income velocity. Part of the rise is cash payments becoming visible for the first time, and the available data cannot separate the two.

Finding 4. Both shocks speeded up adoption, but only one produced a lasting change.

Digital transactions rose 56 per cent in the seven months after demonetisation [24]. But about 98.96 per cent of the withdrawn currency came back to the banking system [23], and cash use recovered a great deal. The pandemic produced 78 per cent growth in FY 2020-21 and then 106 per cent in FY 2021-22, which is faster after the shock than during it. Growth continued at 82 and 57 per cent in the next two years (Data Set 6). The difference seems to be how long each lasted. The pandemic went on long enough for digital payment to become a habit.

Finding 5. UPI has helped financial inclusion in ways that can be measured.

The RBI Financial Inclusion Index rose from 64.2 to 67.0 between March 2024 and March 2025, improving in access, usage and quality [7]. The gain in usage matters most, because access had already been largely achieved. The share of Indian women holding inactive accounts fell from a third in 2021 to 18 per cent by 2024 [21].

Finding 6. There is believable evidence of an effect on real activity, but it is not proof.

Of the small shopkeepers surveyed for the Department of Financial Services, 94 per cent had adopted UPI and 57 per cent said their sales had risen afterwards [15]. These are reported figures that cannot be audited, and a shopkeeper may credit a payment method for a gain that had another cause. Even so, it is the most direct evidence available that the effect reaches output and not just the way payments are made. Research from other countries points the same way. The BIS finds that launching a fast payment system increases the use of digital finance, most strongly in lower-income countries [20].

Finding 7. The gains are shared unevenly, and two serious risks come with them.

In August 2025 rural areas had 46.73 internet subscribers per 100 people against 113.83 in urban areas [22]. That sets a ceiling on how inclusive a system needing a network can be. Cybersecurity incidents rose from 10.29 lakh in 2022 to 22.68 lakh in 2024 [28]. And about 75 per cent of users in one survey said they spent more because digital money felt less real [25]. The same ease that raises velocity also removes a restraint that cash used to provide by accident.

5.2 Suggestions

Six suggestions follow from these findings. Each one addresses a problem the evidence actually shows, rather than a general wish.

17. Treat rural connectivity as payments policy. Data Set 8 shows that what limits rural digital payment is network access, not shop acceptance. About 5.45 crore payment points have been set up in tier-3 to tier-6 centres [14], yet fewer than half of rural people are connected. Extending broadband would therefore do more good now than paying for more acceptance infrastructure.
18. Spend on fraud awareness on the same scale as the system itself. UPI's technical defences are strong: the app is tied to one device, a PIN is required, and machine learning watches for suspicious payments [28]. The weak point is the user's judgement. Awareness campaigns should be treated

as essential infrastructure and delivered in regional languages, inside the payment apps, at the moment of paying.

19. Build spending-awareness features into payment apps. Since most surveyed users say they spend more because digital money feels less real [25], apps could offer optional monthly spending summaries and limits that users set for themselves. This deals with a real cost without taking away anyone's choice.
20. Publish payment data broken down by region. The biggest limitation I faced was that UPI data is not published by state or district, which made it impossible to study regional differences. Releasing this data, with names removed, would allow research that national totals cannot support.
21. Reduce single points of failure. As cash disappears, a breakdown stops being an inconvenience and starts stopping trade. Offline payment modes and modes that work on basic phones deserve continued investment, specifically so the system can survive failures and not only to include more people.
22. Review the incentive scheme openly. Free payment is paid for by the Government rather than by shopkeepers [14], [15]. Whether the gains in recorded activity, inclusion and tax collection are worth that spending is a question that can be answered with evidence, and it should be reviewed publicly from time to time, because the answer decides whether free payment can continue.

6. Conclusion

This project asked whether the growth of UPI has changed the speed at which money circulates in India, and whether any effect on real economic activity can be found in published data. The two questions deserve different answers.

On circulation, the evidence is strong. The value settled over UPI rose from roughly half of nominal GDP in FY 2022-23 to about nine-tenths of it in FY 2025-26. Over the decade the number of transactions rose thirteen times while the average payment shrank by nearly a third. Read through Fisher's equation, that is a large rise in the number of transactions supported by a given stock of money. Read through the Cambridge version, it is a fall in the balance people need to hold against their spending. Both readings describe a rise in the transactions velocity of money. The qualification, stated plainly in Data Set 4, is that part of the measured rise is cash payments moving into a system that counts them. Those payments were always real; they were simply invisible before.

On economic activity the evidence points in one direction but does not prove the case, and this project does not claim more. That 57 per cent of small shopkeepers report higher sales after adopting digital payment [15] is the most direct sign available that the effect reaches output. The BIS finding that fast payment systems raise digital finance use most in lower-income countries [20], and the IMF finding of faster growth in districts where interoperability helped most [18], both point the same way. None of this separates UPI's own effect from the simultaneous spread of smartphones, internet access, bank accounts and higher incomes over the same decade. A study based on published national figures cannot make that separation, and it would be dishonest to pretend otherwise.

What can be said without qualification is that UPI has changed how India pays. It did so by solving a problem cards and wallets never could, which was making the very small payment worth making. The average payment fell from ₹1,838 to ₹1,300 while volume rose more than ten times, and it did so despite inflation pushing the other way. That is the mark of a system reaching transactions it never served before. The tea, the vegetables, the auto fare: these are the payments UPI added, and they are why the national figures look the way they do.

The case for greater efficiency is therefore genuine. Money that does not have to be withdrawn, carried and deposited again is money available for use. A shopkeeper paid instantly can buy new stock sooner. A transaction that leaves a record can later become the basis of a loan. These are real gains and they fit everything found in Chapter 4.

Two problems remain, and neither is minor. The first is unequal access. With rural internet reaching less than half the urban rate, a payment system that needs connectivity cannot include more people than the network does, and the people left out are those for whom inclusion would matter most. The second is security, where the rise in cyber incidents tracks the growth of the system, and where the weak point is the user rather than the technology.

One last observation is worth making. India built this system as public infrastructure and chose to make it free to use, paying the cost from government funds rather than charging shopkeepers. Almost every similar system abroad chose the opposite. The adoption described in this report is, in large part, a result of that decision. That suggests the most important thing about UPI may not be its technology at all, but the view of payment as a public service that the technology was built to serve.

Glossary of Key Terms

Every technical term used in this report is defined below, in the sense in which the report uses it. Terms are listed alphabetically.

Term	Meaning as used in this report
Average ticket size	The average value of a single transaction, obtained by dividing total transaction value by total transaction volume. Derived by the author in Table 2; not published by any source.
Base year revision	A periodic updating of the reference year used to compute national income. India moved from a 2011-12 base to a 2022-23 base during the study period, so GDP figures either side of the change are not perfectly comparable.
CAGR	Compound Annual Growth Rate — the constant yearly rate at which a quantity would have to grow to move from its starting value to its ending value over the period.
Cambridge cash-balance approach	The formulation $M = k \times P \times Y$, which views money demand as the fraction of income people choose to hold as money. Velocity is the reciprocal of that fraction, so $V = 1 \div k$.
Currency in Circulation (CIC)	The total value of banknotes and coins in the hands of the public and in bank tills. India's CIC-to-GDP ratio is high by international standards, and reducing it is a stated aim of RBI policy.
Demonetisation	The withdrawal of legal tender status from ₹500 and ₹1,000 banknotes on 9 November 2016, affecting ₹15.4 trillion, or 86.9 per cent of the value of notes then in circulation.
Digital public infrastructure	Open, interoperable digital systems that support society-wide public and private services. The Bank for International Settlements treats fast payment systems such as UPI as the archetype.
Equation of exchange	Fisher's identity $M \times V = P \times T$, relating the money stock and its velocity to the price level and the volume of transactions. It is true by construction rather than by assumption.
Fast payment system	A retail payment system settling in or near real time, available continuously. UPI is the largest such system in the world by transaction volume.
Financial inclusion	Access to useful and affordable financial services. Measured in India by the RBI's Financial Inclusion Index across three sub-indices: access, usage and quality.
Formalisation	The movement of economic activity from the unrecorded to

Term	Meaning as used in this report
	the recorded economy. A digital payment leaves a record; records accumulate into a credit history.
Income velocity	The number of times a unit of money is used to buy newly produced output, computed as nominal national income divided by the money stock. The conservative measure of velocity, and the one that moves slowly.
Interoperability	The property that a user of one application or bank can transact with a user of any other. This is the design feature the IMF identifies as the source of UPI's adoption.
k (cash-balance ratio)	The fraction of nominal income that people choose to hold as money. A fall in k is arithmetically the same statement as a rise in velocity.
Merchant Discount Rate (MDR)	The fee a merchant pays on each digital transaction. It is zero for UPI transactions up to ₹2,000 for small merchants, with the Government paying an incentive of 0.15 per cent of transaction value in its place.
Natural experiment	A real-world event that changes conditions sharply and without the researcher's intervention, allowing before-and-after comparison. Demonetisation is treated as one in Data Set 5.
Nominal GDP	The value of national output measured at current prices, so it rises with both output and inflation. Used as the denominator in Table 4.
NPCI	The National Payments Corporation of India — the not-for-profit umbrella organisation that built and operates UPI, sets its standards and settles positions between member banks.
P2M and P2P	Person-to-merchant and person-to-person transactions. P2M now forms 63 per cent of UPI volume, and 86 per cent of P2M payments are below ₹500.
PIDF	The Payments Infrastructure Development Fund, which subsidises the deployment of acceptance infrastructure — QR codes and terminals — in tier-3 to tier-6 centres.
RBI Digital Payments Index	A composite RBI index measuring the extent of payment digitalisation, with March 2018 set equal to 100. It stood at 493.22 in March 2025.
Real-time settlement	Settlement of a payment within seconds of it being initiated, as against batch processing at fixed intervals.
Secondary data	Data collected and published by someone other than the researcher. This entire study rests on secondary data, which is stated as its first limitation.
Structural break	A point at which the statistical relationship between

Term	Meaning as used in this report
	variables changes, usually because of a shock or a policy change. COVID-19 is identified as one in the literature reviewed in Section 3.6.
Transactions velocity	The number of times a unit of money is used in any payment, including transfers that are not purchases of output. Payment systems act on this measure directly, and it can move much faster than income velocity.
UPI	The Unified Payments Interface — a common standard, launched in 2016, allowing any bank account to be addressed by any application through a single virtual payment address.
Velocity of money	The average number of times one unit of currency is used to purchase goods and services in a given period. A behavioural property, not a physical one.
Virtual Payment Address	A short identifier that stands in for a bank account number and branch code when making a UPI payment.

A note on one ambiguity: the abbreviation DPI is used in two senses in the payments literature — "digital public infrastructure" and the RBI's "Digital Payments Index". This report spells both out in full wherever they appear.

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